L-P-395C
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SUPERSEDING
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November 13, 1969

FEDERAL SPECIFICATION

PLASTIC MOLDING AND EXTRUSION MATERIAL

NYLON, GLASS FIBER REINFORCED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers glass fiber reinforced nylon (polyamide) compounds suitable for injection molding and extrusion.

1.2 Classification.

1.2.1 Types, classes, and electrical grade. The reinforced nylon materials shall be of the following types, grades, and classes, as specified (see 6.2 and 6.5).

Type I - Nominal 30 percent glass.
  Grade A - Electrical
  Grade B - General purpose

Type II - Nominal 40 percent glass
  Grade A - Electrical
  Grade B - General purpose

Class 1[1/] - See table I
Class 2[1/] - See table I
Class 3[1/] - See table I

[1/] Difference in classes cannot be defined adequately without use of table I. Use of Izod impact test values for classification may be misleading (see footnote 2 to table I).

FSC 9330

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-D-723 - Drums, Fiber
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agency).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials (ASTM) Standards

D 149 - Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies.
D 150 - A. C. Loss Characteristics and Dielectric Constant (Permittivity) of Solid Electrical Insulating Materials.
D 256 - Impact Resistance of Plastics and Electrical Insulating Materials.
D 570 - Water Absorption of Plastics.
D 638 - Tensile Properties of Plastics
D 789 - Nylon Injection Molding and Extrusion Materials.
D 790 - Flexural Properties of Plastics
D 792 - Specific Gravity and Density of Plastics by Displacement.
D 1897 - Injection Molding of Specimens of Thermoplastic Molding and Extrusion Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606)

3. REQUIREMENTS

3.1 Material. The material shall consist of homogeneous glass fiber reinforced virgin nylon injection molding and extrusion compound with or without the addition of additives, stabilizers, or modifiers. The nylons covered are 6/6, 6, 6/9, 6/10 and 6/12. The allowable range in glass content for type 1 material shall be 30 to 35 percent. The allowable range in glass content for type II material shall be 40 to 45 percent, with the exception that nylon 6/9, 6/10 and 6/12 shall have an allowable range of 40 to 50 percent glass fiber.

3.2 Property values. When tested as specified in the applicable procedure of 4.3, grade B material shall conform to the property values shown in table I. When grade A (electrical grade) material is specified, it shall conform to the property values specified in tables I and II.

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TABLE I. Property values (nonelectrical) for type I and type II, grade A and grade B material[1/]

Material	Tensile strength, p.s.i., min.	Flexural modulus 10^6 p.s.i., min.	Izod impact strength[2/] ft.-lb./in., min.	Melting point range	Specific gravity min.	Specific gravity max.	Water absorption, 24 hours, percent, max.
Type I							
Nylon 6/6							
Class 1	20,000	1.0	2.5	249° to 266°C	1.34	1.42	1.10
Class 2	24,000	1.2	2.0	480° to 510°F	1.35	1.42	
Class 3	24,000	1.2	1.5		1.35	1.42	
Nylon 6							
Class 1	21,000	1.0	2.8	204° to 221°C	1.35	1.42	1.20
Class 2	22,000	1.2	2.0	400° to 430°F			
Class 3	22,000	1.2	1.5				
Nylon 6/10 and 6/12							
Class 1	20,000	0.8	3.0	199° to 219°C	1.31	1.38	0.30
Class 2	20,000	1.0	2.3	390° to 425°F			
Class 3	20,000	1.0	1.8				
Nylon 6/9							
Class 1	20,000	0.8	3.0	199° to 219°C	1.31	1.38	0.40
Class 2	20,000	1.0	2.3	390° to 425°F			
Class 3	20,000	1.0	1.8				
Type II							
Nylon 6/6							
Class 1	30,000	1.3	3.3	249° to 266°C	1.46	1.53	0.90
Class 2	30,000	1.5	2.5	480° to 510°F			
Class 3	30,000	1.5	2.0				
Nylon 6							
Class 1	29,000	1.3	4.0	204° to 221°C	1.46	1.53	1.00
Class 2	29,000	1.5	3.2	400° to 430°F			
Class 3	29,000	1.5	2.5				
Nylon 6/10 and 6/12							
Class 1	28,000	1.2	4.5	199° to 219°C	1.40	1.53	0.25
Class 2	28,000	1.3	3.0	390° to 425°F			
Class 3	25,000	1.3	2.5				
Nylon 6/9							
Class 1	28,000	1.2	4.5	199° to 219°C	1.40	1.53	0.35
Class 2	28,000	1.3	3.0	390° to 425°F			
Class 3	25,000	1.3	2.5				

[1]An option is permitted in use of long glass fibers or short glass fibers (see 4.2.1).
[2]Other impact tests, including tensile impact and drop weight impact, may show different relationships between impact behavior of these materials. Izod impact is included for material identification only and does not predict nor correlate with performance of parts molded of these materials.

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TABLE II. Electrical property values (for grade A material only)

Property	Types I and II, classes 1, 2 and 3, nylons 6/6, 6, 6/9, 6/10 and 6/12
Dielectric strength, short time test on 1/8 in. thick specimen, volts per mil, min.	375
Dielectric constant at 1 megahertz, max.	4.2
Dissipation factor at 1 megahertz, max.	0.025

3.3 Form. The form of the material shall be as specified by the procuring agency (see 6.2 and 6.3). When a particular particle size is specified, the limits and method of testing shall be as specified by the procuring agency (see 6.2).

3.4 Color. The color shall be natural. If color is specified, the color property values, and methods of inspection for color shall be as agreed upon between supplier and procuring agency (see 6.2).

3.5 Workmanship and uniformity. The material shall be uniform in form and color within each container and from container to container. The material shall be free from dirt and other contamination as determined by visual examination.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same nylon and same type, grade, class, and either long or short glass fibers submitted for delivery at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used, were manufactured, inspected, and tested in accordance with the requirements of this specification, and, to the extent specified, of all referenced subsidiary specifications and standards. In the event of conflict, this specification shall govern. A supplier's certificate of compliance with 3.1, and that the color is natural (see 3.4), shall be furnished. The certificate shall state the nylon as 6/6, 6, 6/9, 6/10, 6/12 or modifications thereof (see 3.1), as applicable. The use of long glass fibers or short glass fibers shall be specified in the certificate.

4.2.2 Inspection of material.

4.2.2.1 Examination of the material. Examination of the material shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQL's) set forth below. The lot size for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of 100 pounds for examination in 4.2.2.1.1, and in units of shipping containers for examination in 4.2.2.1.2.

4.2.2.1.1 Examination of the material for defects in appearance and workmanship. The sample unit for this examination specified in table III shall be approximately one pound.

TABLE III. Examination of the material for defects in appearance and workmanship.

Examine	Defect
Appearance and workmanship	Form not as specified Color not as specified Form not uniform Color not uniform Not clean, presence of foreign material

4.2.2.1.2 Examination of the preparation for delivery requirements. An examination shall be made in accordance with table IV to determine that packaging, packing, and marking comply with section 5 requirements. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

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TABLE IV. Examination of the preparation for delivery

Examine	Defect
Packing	Not level specified; not in accordance with contract requirements. Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability. Inadequate application of components such as: incomplete closure of polyethylene liners.
Quantity of material	Less than specified or indicated quantity.
Weight	Gross weight exceeds specified requirements.
Markings	Exterior markings omitted, illegible, incorrect, incomplete, of improper size, location, sequence or method of application, or not in accordance with contract requirements.

4.2.2.1.3 Inspection levels and acceptable quality levels (AQL's) for examinations. The inspection levels for determining the sample size and the acceptable quality level (AQL) expressed as defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1.1	II	2.5
4.2.2.1.2	S-2	2.5

4.2.2.2 Classification of tests. All tests under this specification shall be classified as lot acceptance tests. Lot acceptance tests shall be made on each lot of material and, in conjunction with the above examination, shall be the basis of acceptance or rejection of the lot.

4.2.3 Testing. The material shall be tested for the characteristics listed in table I and table II (for electrical grade material only), in accordance with the test methods specified herein. The lot size, for the purpose of determining the sample size for testing shall be expressed in units of 200 pounds. The sample unit shall consist of sufficient material to prepare all required specimens. The inspection level shall be S-1 with an acceptance number of 0. The results for each test shall be averaged results of the specimens, unless only one specimen or determination is specified for testing.

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4.3 Test methods.

4.3.1 Specimen preparation and conditioning. Unless otherwise specified, specimens shall be prepared by injection molding in accordance with ASTM D 1897 using stock and mold temperatures recommended by the manufacturer (see 6.4). Immediately after removal from the mold, specimens shall be sealed in containers which are impermeable to water vapor, such as screwcap glass containers with taped closures. All specimens shall be conditioned for a minimum of 24 hours in the closed containers maintained at 23° ± 2°C. (73.4° ± 3.6°F.).

4.3.2 Specimen testing. Individual specimens shall not be removed from closed containers until ready for testing. Testing shall be at 23° ± 2°C. (73.4° ± 3.6°F.) and 50 ± 5 percent relative humidity.

4.3.3 Tensile strength. Three specimens shall be tested in accordance with ASTM D 638, using speed B. Specimens shall be 1/8 inch thick and other dimensions shall conform to type I of figure 1 of ASTM D 638.

4.3.4 Flexural modulus. Three specimens shall be tested in accordance with procedure A of ASTM D 790. Specimens shall be 1/4 inch thick and, other dimensions and crosshead speed, shall conform to table I of ASTM D 790. The tangent method shall be used for calculations.

4.3.5 Izod impact strength. Three specimens shall be tested in accordance with ASTM D 256. Specimens shall be 1/4 inch thick and tested as a single piece and not a composite of thinner sections.

4.3.6 Melting point. One determination for melting point shall be made on the molding or extrusion compound in accordance with ASTM D 789. Conditioning and conditions of testing for the compound shall be in accordance with that specified in 4.3.1 and 4.3.2 for specimen testing.

4.3.7 Specific gravity. One specimen shall be tested in accordance with method A-1 or A-2 of ASTM D 792.

4.3.8 Water absorption. Three specimens shall be tested in accordance with ASTM D 570. Each specimen shall be 2 inches by 1/8 inch disc with dimensional tolerances specified in ASTM D 570.

4.3.9 Dielectric strength (applicable only to grade A material). The dielectric strength of 5 specimens, each 4 inches in diameter and 1/8 inch thick, shall be determined by the "short-time test" of ASTM D 149.

4.3.10 Dielectric constant and dissipation factor (applicable only to grade A material). The dielectric constant and dissipation factor shall be determined on 5 specimens, each 4 inches in diameter and 1/8 inch thick, in accordance with ASTM D 150 using a frequency of one megahertz.

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5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified (see 6.2), the material in the quantity specified (see 6.2), shall be packed in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type II, grade A, or type III, grade A in quantities of 150 pounds maximum.

(b) Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds maximum.

(c) Five gallon metal cans conforming to PPP-C-96, type V.

Insofar as practical, containers shall be of uniform shape and size, with minumum cube and tare consistent with the protection required. Containers shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with an 0.004 inch thick polyethylene liner either properly heat sealed or properly tied.

5.1.2 Level B. Unless otherwise specified (see 6.2), the material, in the quantity specified (see 6.2), shall be in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type I, grade A in quantities of 150 pounds maximum.

(b) Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds maximum.

(c) Five gallon metal cans conforming to PPP-C-96, type V.

Insofar as practical, containers shall be of uniform shape and size with minimum cube and tare consistent with the protection required. Containers shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with an 0.004 inch thick polyethylene liner properly heat sealed.

5.1.3 Level C. Packing shall be in accordance with commercial practice adequate to ensure acceptance and delivery by the carrier for the mode of transportation employed. Containers shall comply with the requirements of the Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

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5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking specified in the contract or order (see 6.2), shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military requirements. In addition to any special marking specified in the contract or order (see 6.2), shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Glass fiber reinforced nylons may be used where strength, stiffness, abrasion and impact resistance, chemical resistance, durability, and low coefficient of friction are required. The glass reinforced nylons offer increased resistance to deformation, lower creep, and increased stiffness at elevated temperatures when compared with unfilled nylons.

Types I, and II material represent different glass contents and are intended for applications where a particular glass content is optimum.

Glass reinforced nylon 6 is intended for use where high impact resistance along with high strength and stiffness is required. Glass reinforced nylon 6/6 is intended for use where high strength and stiffness over a wide range of temperatures, as well as the highest heat deflection temperature and melting point are required. Glass reinforced nylon 6/9, 6/10, and 6/12 are intended for use where low moisture absorption, high impact strength, excellent dimensional stability, and better electrical properties are required.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, grade, and class of material required (see 1.2, 3.1, and 3.2).
(c) Nylon as 6/6, 6, 6/9, 6/10, or 6/12, as required (see 3.1 and 4.2.1).
(d) Form, if required (see 3.3).
(e) Color, if required, and agreement with supplier (see 3.4).
(f) Level of packing required (see 5.1).
(g) If other containers are required (see 5.1.1 and 5.1.2)
(h) Quantity in shipping container (see 5.1.1 and 5.1.2).
(i) Special markings required (see 5.2.1 and 5.2.2)

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6.3 Fabricator choice of form. The fabricator of molded parts or extruded shapes is expected to recommend the material form best suited to his equipment (see 3.3).

6.4 Injection molding of specimens. Stock and mold temperatures recommended by the manufacturer should be used for molding. For attainment of optimum physical properties, use of a reciprocating screw type of injection machine is recommended.

6.5 Cross index. Table V shows the classification of material covered by this specification and the corresponding classification of material specified in L-P-395A and L-P-395B.

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TABLE V. Classification in this document and corresponding classification in L-P-395B and L-P-395A

L-P-395C	L-P-395B	L-P-395A
Types I and II, class 1 (for nylons 6/6 and 6, and 6/10)	Types I and II, class 1 (for nylons 6/6, 6, and 6/10	Types I and II (for nylons 6/6, 6, and 6/10, respectively)
Deleted class 4	Class 4 for all types	None
Types I and II, classes 2 and 3 (for nylons 6/6 and 6, and 6/10)	Types I and II, classes 2, 3, and 4 (for nylons 6/6, 6, and 6/10)	None
Grade A	Grade A	Electrical grade
Grade B	Grade B	General purpose and low water absorption
Nylon 6/9	None	None
Nylon 6/12	Nylon 6/12, all types and classes	None

MILITARY CUSTODIANS:

Army - MR
Navy - SH
Air Force - 11

Review activities:

Army - AV, MI, EL, GL, MD
Navy - SH
Air Force - 84
DSA - GS

User activities:

Army - ME
Navy - AS, OS

Preparing activity:

Army - MR

CIVIL AGENCIES INTEREST:

GSA - FSS
MSF
FEC

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